The Cost of a Bounded Vortex: A Compact-Support Artifact in Tropical Cyclone β-Drift and Its Removal

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Abstract. A companion study characterized a persistent bias in an idealized tropical-cyclone model's self-propagation—β-drift of canonical magnitude aimed nearly due poleward—and showed it to be subdominant to environmental steering at landfall, without identifying its cause. Here we identify the cause, remove it, and measure what its removal is worth. The bias is produced by the compact-support taper used to bound the initial vortex: truncating the outer wind removes the ambient flow that phase-locks the β-gyres, leaving them in free cyclonic precession. Replacing the taper with a smooth Gaussian envelope restores the lock—the gyre orientation is set within twelve hours and held—and yields β-drift that is canonical in magnitude and aim, robust across envelope scale, intensity-invariant in direction, and null-clean on an f-plane. In a six-storm reanalysis-steered A/B test conducted under predictions registered before each run, the corrected drift shifted every storm's landfall cross-track westward with the predicted sign, at approximately one-third the linearly projected magnitude—an attenuation that a direct test attributes not to steering feedback but to the gap between mature-testbed drift and in-transit drift (intensity scaling and gyre spin-up)—quantifying why self-propagation biases are subdominant at landfall, and why landfall error is an unreliable instrument for reading them. A persistence baseline and a steering-only tracer, run under the same registered-prediction discipline, locate the skill (steering sets cross-track, the vortex's drift sets timing, and the model beats persistence on all six landfalls), and a three-storm blind extension makes the sign record nine for nine while reproducing the mechanism's per-storm predictions on straight movers—the two storms that curve sharply into landfall under-transmit alike, a reproducible residual we characterize rather than explain. The correction's own cost is characterized: the envelope delays dry barotropic re-intensification by 8–12 hours by weakening mid-radius Ekman inflow.

Significance Statement. Hurricane track models must represent not only how storms are steered by their environment but how they propel themselves. We show that a common and innocent-looking modeling choice—truncating a simulated storm's outer winds at a fixed radius so the vortex fits a bounded domain—silently disables the mechanism by which the storm's self-propagation finds its correct direction, leaving it aimed nearly due poleward. Replacing the truncation with a smooth envelope restores textbook behavior at the cost of one line of code. On six historical hurricanes the correction moved every landfall in the predicted direction, but at only one-third the predicted distance—a measured attenuation that explains why errors of this kind hide at landfall. Any model that bounds its storms this way can check for the artifact in an afternoon.

1 Introduction

The companion paper (in preparation) reported two facts about the model considered here and declined to bridge them: a characterized error—the simulated β-drift, canonical in magnitude, aims nearly due poleward, its westward component a fraction of the canonical value—and blind track skill across six historical landfalls under a configuration with no landfall-tuned parameter. A projection test showed the error could not be the dominant source of landfall cross-track error, and the paper left its *cause* as an open question with three named candidates: the compact-support cutoff that bounds the initial vortex's outer wind, the reduced barotropic configuration, and the shape of the taper ramp.

This paper answers the question and follows the answer to its consequences. The cause is the first candidate. The compact-support taper—introduced, reasonably, to give a periodic domain a vortex of finite size—removes the outer circulation that the β-gyres require to phase-lock, and an unlocked gyre pair precesses freely past its equilibrium orientation. A smooth envelope that bounds the vortex without truncating it restores canonical behavior entirely. We then ask what the fix is worth on real tracks, in a six-storm A/B test against the companion paper's record, with per-storm predictions registered before any run; the answer—every storm moves the predicted direction, at a third of the predicted distance—measures a quantity of independent interest, the factor by which landfall displacement under-reads a self-propagation change, whose mechanism we then identify by direct test. Finally, because no vortex boundary is free, we characterize the envelope's own cost: it delays the model's dry barotropic re-intensification, and we measure the mechanism of the delay.

Throughout, we follow the companion paper's discipline: predictions registered before runs, controls before claims, and failures reported with the successes. Of thirty-two quantitative predictions registered across the campaign, fifteen were confirmed, ten failed, and seven were partial, unscoreable, or resolved against our stated lean; every failure narrowed the search, and two of the failures (Sections 2 and 5) taught us more than the confirmations did.

2 Candidate Experiments

2.1 Design

The testbed is the companion paper's: a balanced Holland vortex (Holland 1980) on a quiescent β-plane (maximum wind 64 m s^{-1}, radius of maximum wind 75 km, environmental radius 500 km, intensity cap 70 m s^{-1}, 320^2 grid points at $\Delta x = 15.6$ km on a 5000-km domain), with the mature β-drift vector read from the 30–48-h window. Two experimental arms address the three candidates. Arm A varies the outer-wind profile at fixed geometry: three compact-support ramps of increasing smoothness—linear (slope discontinuities at both ends), the production cosine, and a quintic smoothstep (vanishing first and second derivatives)—plus two profiles with no compact support at all, Gaussian envelopes $\exp(-(r/r_d)^2)$ with $r_d = 420$ km (matched to the production profile's wind at 350 km) and 560 km (Fig. 1). One limit of this design is stated up front: because compact support fixes the removed circulation, all three ramps carry an *identical integrated ring strength* and differ only in the sharpness of the ring's edges—so Arm A separates edge sharpness from the cutoff, not the ring's presence from it, and a ring-carrying profile that retains the outer tail is required to finish the attribution (Section 3). Arm C addresses the barotropic-configuration candidate with a baroclinicity ladder: the balanced warm core retained and buoyancy activated, with thermal relaxation at the production timescale, at a sixfold longer one, and disabled.

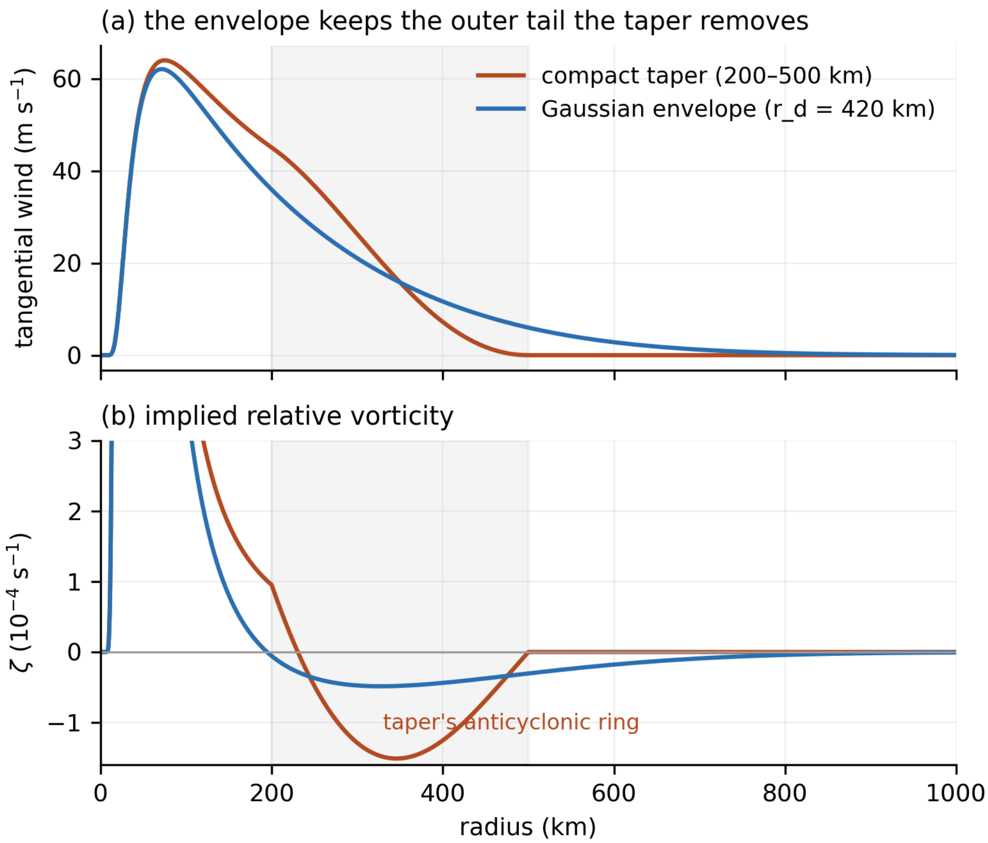

Fig. 1. The two outer boundary conditions. (a) Tangential-wind profiles of the production compact taper (cosine ramp to zero over 200–500 km; band shaded) and the Gaussian envelope (r_d = 420 km), whose wind matches the taper's at 350 km by construction: the envelope keeps the outer tail the taper removes. (b) The implied relative vorticity: the taper concentrates a strong anticyclonic ring inside the ramp band; the envelope spreads a weaker, broader negative lobe and retains far-field circulation.

The honest metric, throughout, is the westward drift component—in the biased model it is the deficient quantity: ≈ 0.4 m s^{-1} against the $\sim$ 1–2 m s^{-1} implied by the canonical magnitude and direction bands (1–3 m s^{-1} toward the northwest quadrant; Chan 2005)—and, unlike the heading, it is not confounded by intensity—and every reading is reported against the equilibrated maximum wind, because an earlier probe in the companion study found aim rotations that were artifacts of vortex collapse.

2.2 Arm A: The Cutoff, Not the Ramp

Table 1. Arm A: mature (30–48 h) testbed β-drift by outer-wind profile. Heading is measured clockwise from due north; "westward" is the westward component of the drift. $V_{\max}$ end is the equilibrated maximum wind.

Profile	Drift (m s^{-1})	Heading	Westward (m s^{-1})	$V_{\max}$ end (m s^{-1})
linear ramp, 200–500 km	2.19	348°	+0.47	42.3
cosine ramp (production)	2.49	350°	+0.42	42.2
quintic ramp	2.64	352°	+0.35	43.7
Gaussian, $r_d = 420$ km	2.30	**329°**	**+1.20**	39.7
Gaussian, $r_d = 560$ km	2.85	**326°**	**+1.60**	41.2

Ramp form does nothing: across a threefold change in ring sharpness the heading spans 4° and the westward component 0.12 m s^{-1}. The cutoff does everything: both envelopes rotate the drift 21–24° into the canonical northwest band (Chan 2005) and recover a westward component of 1.2–1.6 m s^{-1}—three to four times the compact family's—at essentially preserved intensity (Table 1). Three confounds are excluded by construction. Circulation size cannot be the driver, because the two families respond to size with *opposite* signs (broader compact vortices drift more poleward; broader envelopes drift more westward). Reduced mid-radius wind cannot be, because the broader envelope carries more mid-radius wind than the narrower one yet is more canonical, and the companion study's onset-radius sweep—which reduces mid-radius wind directly—never escaped its 343° floor. And intensity cannot be, because the equilibrated winds differ by at most 2.5 m s^{-1}.

2.3 Arm C, Reported Honestly

The baroclinicity ladder's null rungs behaved exactly (the retained-but-passive warm core reproduced the dry control to the tracker's precision), but its live rungs were unreadable as designed: relaxing temperature perturbations toward zero cannot hold a *balanced* warm core, so weakening the relaxation produced not persistent baroclinicity but runaway adiabatic warming—95-K anomalies, far outside the anelastic core's validated regime, with the intensity cap engaged. The corrected design—relaxation toward the initial balanced core—closed the runaway: the held vortex carries a bounded 45-K warm core indefinitely, and a passive-core null reproduces the dry control exactly. It also revealed why this candidate resists clean adjudication: a maintained warm core is an energy source, and the held-baroclinic vortex intensifies to the cap, entangling the aim comparison with the known intensity dependence. Two further compromises are named so the reading is discounted properly: the held 45-K core sits at $\theta'/\bar{\theta} \approx 15\%$, a factor of 1.7 beyond the anelastic system's directly validated range ($\approx 9\%$), and the model's heated secondary circulation is not vertically converged at the production spacing (Section 6)—Arm C is the only buoyancy-active experiment in either paper and inherits both. The reading that survives all three is directional. With live, maintained vertical structure, the drift moves *poleward* of the dry control (356° versus 350°; westward component 0.13 versus 0.42 m s^{-1})—the wrong direction to explain the westward deficit. Whatever maintained baroclinicity does to this core's β-gyres, it does not supply the missing westward propagation: the candidate is exonerated as the cause of the bias by sign, with only the magnitude of its wrong-way nudge left entangled with intensity (Section 6).

3 The Mechanism: Phase-Locking Requires the Outer Flow

The time-resolved traces make the mechanism legible (Fig. 2). Under every compact-support profile the gyre pair reaches canonical *amplitude*—drift speed saturates near 2.5 m s^{-1}—while its *orientation* never locks: the heading precesses cyclonically at $\approx 0.4°$ h^{-1}, through due north and beyond, for as long as we have integrated (60 h). Under the envelopes the orientation is set by t = 12 h, at roughly half the final amplitude, and held to $\pm 2°$ through $t = 48$ h while the amplitude grows to saturation (Fig. 2b).

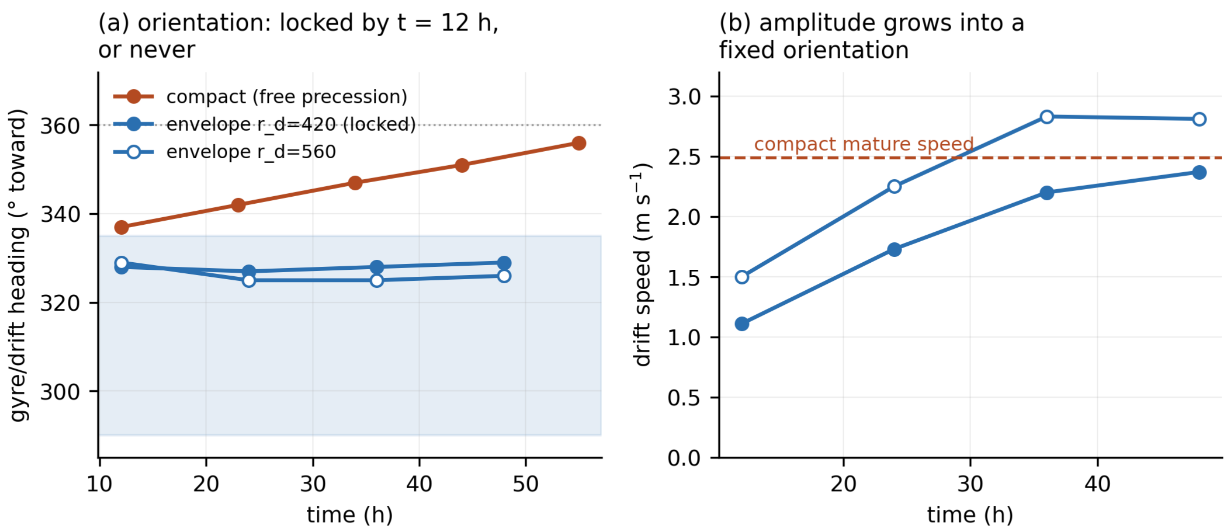

Fig. 2. The phase lock. (a) Gyre/drift heading versus time: the compact profile precesses freely through due north; both envelopes lock by t = 12 h inside the canonical band (shaded) and hold. (b) Drift speed versus time: under the envelopes the amplitude grows into a fixed orientation—orientation is set early, at roughly half the final amplitude, and held while the drift matures.

In the canonical picture, the gyres equilibrate where their generation by the β-effect balances their advection by the vortex circulation (Fiorino and Elsberry 1989; Chan and Williams 1987), and Fiorino and Elsberry located the controlling flow in the outer wind, roughly 300–800 km from center (Fiorino and Elsberry 1989; Smith et al. 1990). The compact taper amputates precisely that band's outer half. What remains cannot advect the gyre pair into its equilibrium orientation, and the asymmetry free-runs at something like a β-Rossby precession rate. The "over-rotation" of the companion paper was therefore never a rotation-rate error; it was the absence of the arresting flow.

What is new here should be stated against a literature that is not. That the outer wind controls β-drift is established: the symmetric tangential wind in the 300–800-km annulus sets the gyres' magnitude and orientation (Fiorino and Elsberry 1989; Smith et al. 1990), an empirical law gives drift direction as a function of profile shape (Smith 1997), and Carr and Elsberry (1997) built an operational beta-and-advection propagation model on compactly supported vortices indexed by their cutoff radius. Prior work, that is, shows the *equilibrium orientation shifting* with outer profile. The compact family here shows something categorically different: no equilibrium at all—amplitude saturates while orientation precesses freely at $\approx 0.4^\circ$ h^{-1}, through due north and beyond, for as long as we have integrated (Fig. 2a). That is a failure mode, not a parameter dependence. How Carr and Elsberry's cutoff vortices nonetheless yield well-defined propagation is a comparison we flag rather than resolve—where their cutoff sits relative to the gyre annulus, and what their closure assumes about equilibration, are the candidate differences—and the afternoon diagnostic of Section 6 applies to their configuration as written.

The recovered drift also changes *kind*, in a way that identifies it as canonical structure rather than a retuned artifact. In the compact family the westward component was fixed (≈ 0.41 m s^{-1} at every intensity) while the northward component scaled with maximum wind—the drift direction rotated with intensity (338° to 351° across a threefold range). In the envelope family the direction is fixed (324–329° across the same range) while the whole vector scales—which is what idealized β-drift theory requires of a structurally set propagation. Calibration of the envelope scale confirms the lock is not a tuning accident: across $r_d = 350$–560 km the heading spans 4° while the magnitude ranges 1.99–2.85 m s^{-1}—aim and size, entangled all the way to the compact family's floor, have decoupled. An f-plane null closes the case: with β removed the envelope vortex drifts 0.002 m s^{-1}, so the envelope manufactures nothing.

4 The Fix and Its Calibration

The production change is one line: the outer boundary condition of the initial vortex, from cosine-to-zero over 200–500 km to a Gaussian envelope. The scale was frozen at $r_d = 420$ km before any storm run, on testbed grounds alone: its wind matches the production taper's at 350 km by construction (making the storm A/B maximally "the same vortex, plus the tail"), its drift magnitude (2.30 m s^{-1}) sits nearest the production control's 2.49, and its westward component (+1.20) lies nearest the canonical value. The envelope's far field is negligible where the domain requires it (0.04 m s^{-1} at 1000 km at storm intensity). How little far field the lock actually needs is worth recording: the narrowest calibrated envelope, $r_d = 350$ km, locks at 330° while carrying 3.2 m s^{-1} at 500 km and 0.4 m s^{-1} at 700—the arresting flow is bought with a few meters per second beyond the taper's own support, which sharpens the exported diagnostic considerably. The no-landfall-tuning discipline of the companion paper thus survives the model change that paper motivated: the new parameter, like the old, has never seen a landfall.

5 Six Storms, and the Transit Attenuation

5.1 The Differential Projection

The companion paper's projection test—which falsified the bias-explains-landfall bridge—has a better-posed differential form: in an A/B of the *same model* under the *same* reanalysis steering (ERA5; Hersbach et al. 2020), the steering errors cancel, and the drift correction $\Delta = (-0.78, -0.49)$ m s^{-1} (east, north), projected through each storm's landfall geometry and transit time, yields a per-storm predicted displacement. (Δ is a $\Delta x = 15.6$-km quantity: the testbed drift magnitude decreases $\approx 8\%$ across a grid doubling with no demonstrated asymptote (companion paper), and the strong-form projections and every ratio built on them inherit that resolution dependence.) We registered those predictions, with two hypotheses for their amplitude: H1, linear accumulation (the strong form); H2, partial absorption by the steering relaxation, which samples the environment at the storm's actual position and therefore pulls a displaced storm back toward the track the environment dictates. Guards were registered alongside: intensity histories within ± 10 m s^{-1} (the envelope's re-intensification effect, Section 5.3, was already suspected), and timing within ±3 h on the direct-landfall storms.

5.2 Results: The Right Sign Six Times, at a Third of the Distance

Every storm shifted westward, as predicted—discovery and test set, fast and slow movers, straight runners and recurvers alike (Table 2, Fig. 3).

Table 2. The six-storm A/B: cross-track error at the observed landfall fix under the control (compact-taper) and envelope configurations, the observed shift, the strong-form (linearly projected) shift, and their ratio. Daggers mark the two intensity-guard exceedances, excluded from the transmission estimate.

Storm	Cross-track, control → envelope (km)	Observed Δ (km)	Strong-form Δ (km)	Ratio
Hugo	+110.2 → +78.7	−31.5	−93	0.34†
Katrina	+124.6 → +78.2	−46.4	−108	0.43
Ivan	+126.3 → +40.2	−86.1	−139	0.62†
Fran	+7.7 → −29.9	−37.6	−81	0.46
Michael	−98.7 → −113.2	−14.5	−73	0.20
Laura	−31.9 → −51.8	−19.9	−74	0.27

The strong form failed; the attenuated form is what the storms show. On the guard-clean storms the transmission ratio—observed shift over linear projection—averages **0.34** (range 0.20–0.46): landfall registered roughly one-third of the self-propagation change (Fig. 3). Section 5.3 tests, and overturns, our registered attribution of that attenuation. Along-track error improved on the poleward movers, where the companion paper's projection test had located the bias's real footprint (Katrina +76.5 → +61.4, Michael +123.8 → +93.4, Laura +37.4 → +30.0 km). The guards did their work: two storms—Hugo and Ivan, the two with strong re-intensification phases—ran 14 and 25 m s^{-1} weaker under the envelope, so their larger apparent gains are entangled with weakened drift magnitude and are flagged, not claimed. The honest skill accounting is correspondingly modest: six-storm cross-track RMS falls from 95 to 71 km in aggregate but only from 81 to 75 km on the guard-clean four; along-track RMS falls from 121 to 101 km.

A baseline, added when external readers asked what those numbers are worth, calibrates them. Under predictions registered before the calculation, a persistence forecast—each storm's observed motion at initialization, extrapolated over its transit—was scored with the same landfall-fix decomposition. It loses to the control configuration on landfall position for all six storms (mean total error 225 km against 132; margins from ×1.1 on Ivan to ×4.7 on Laura), with the deficit concentrated along-track: every storm in this sample accelerated poleward, and persistence cannot know that. Its cross-track RMS alone, however, is 117 km against the control's 95—under near-true steering, even a no-skill extrapolation is not catastrophically wrong about the cross-track. The registered expectation of a more-than-twofold cross-track margin failed, and the failure sharpens the claim: the model's edge over no-skill is landfall position and timing; cross-track placement is largely the steering's, which is the companion paper's attribution restated as a measurement.

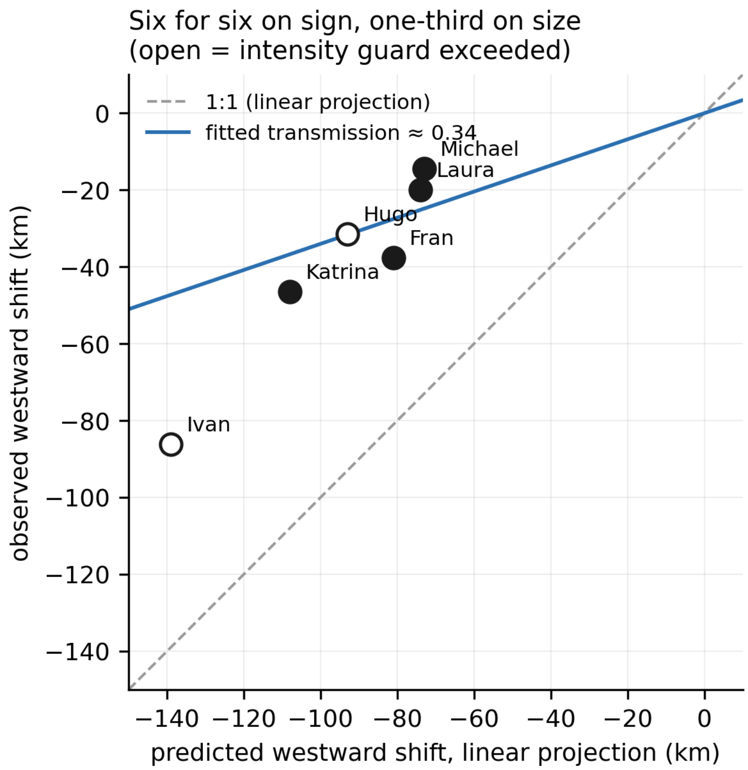

Fig. 3. Observed westward cross-track shift versus the strong-form (linear) projection for the six-storm A/B. Every storm moves west, as predicted; the drawn line is the guard-clean mean transmission, 0.34 (a through-origin fit gives 0.37; dashed line: 1:1). Open symbols mark the two intensity-guard exceedances, excluded from the estimate.

5.3 What the Attenuation Is—and Is Not—and the Correction's Own Cost

The transmission ratio is this paper's most exportable number, and our first attribution of it was wrong. We registered the natural hypothesis (H2): the relaxation samples the environment at the storm's own position, so a displaced storm feels a restoring pull. The direct test severs that loop—steering sampled along the observed track, so that both profiles receive an identical forcing series—and the A/B delta barely moved: −43 km against −46 with the feedback live. Whatever attenuates the drift correction, it is not the position feedback (bounded below $\sim 8\%$ for this pair). The accounting that fits is simpler and bias-side: the testbed characterizes the drift at mature equilibrium and full strength, while the real storm runs weaker (the envelope-family drift scales with intensity—a factor ≈ 0.64 at Katrina's) and spends its first half-day spinning the gyres up (≈ 0.65 of the transit)—the product, ≈ 0.42, against Katrina's observed 0.43. Landfall reads self-propagation error at one-third strength not because the architecture pulls the storm back, but because a mature-testbed drift vector overstates what the drift does during a real transit. The caution for attribution studies survives unchanged and becomes more general, not less: landfall error under-reads self-propagation bias by roughly a factor of three for reasons intrinsic to how drift develops over a transit—intensity history and gyre spin-up—not to any particular steering architecture. The number should travel; our first attribution of it should not.

The accounting was then asked to do something harder than match a mean. Made per-storm—the family drift-versus-intensity laws evaluated on each run's own intensity history, under the testbed spin-up curve—it predicts a transmission ratio for each storm, registered before computation. Three of the four guard-clean storms land within ± 0.11 (Katrina 0.52 predicted, 0.43 observed; Fran 0.43, 0.46; Laura 0.38, 0.27); the guard-clean means, 0.44 predicted against 0.34 observed, put the accounting 26% high in aggregate, an overshoot driven entirely by Michael. One bookkeeping rule is worth stating in a paper about compensating errors: a storm the guard excludes from an estimate never reappears in that estimate's validation target. Ivan is in no observed average above; the all-six aggregate (0.42) and the guard-clean aggregate (0.35) are reported in the record for completeness, and the number this paper exports is the guard-clean mean of ratios, 0.34. It also closes the books on a flagged storm: Ivan's outsized 0.62, which the intensity guard excluded, is predicted at 0.67 by the same accounting—the weakened envelope run's poleward deficit projects into cross-track at Ivan's heading, so the guard removed exactly the entanglement the decomposition now quantifies. The residual is Michael: 0.45 predicted against 0.20 observed on cross-track, while its along-track transmission matches the prediction (0.46 observed, 0.38 predicted). The outlier is axis-specific—Michael is the one storm whose landfall heading makes the cross axis nearly zonal—and we record it as unresolved rather than explained.

An independent instrument corroborates the aggregate. A steering-only tracer—a point advected by the sampled annulus DLM, no vortex—separates what the steering delivers from what the vortex adds: the tracer reproduces the six-storm cross-track record nearly as well as the full model (105 km RMS against 95) but arrives late everywhere, the model running ahead of it along-track on all six storms. The model-minus-tracer displacement at landfall falls within a factor of two of the attenuated drift footprint on all four guard-clean storms (1.2–1.7$\times$). Two instruments that share no machinery—an A/B under identical steering, and a vortex-free integration of the steering itself—land on the same transmission accounting.

The intensity-guard violations are not noise; they are the fix's price, and we measured it. In the model's emergent-intensification testbed (β-plane with a steering ramp, the configuration in which this dry model demonstrably re-intensifies), the envelope does not suppress the re-intensification—it postpones it, by 8–12 h, to the same capped equilibrium (onset of intensification at 24 h under the compact profile, 32 h under the envelope; matched-time differences reach 30 m s^{-1} mid-run and collapse to 3 by the end) (Fig. 4a). A moving-frame angular-momentum budget locates the mechanism: in the pre-onset window the boundary-layer mass inflow at 300 km runs 2.5–3$\times$ stronger under the compact profile (Fig. 4b)—Ekman convergence scales with the surface wind the profile places at 200–400 km—while the candidate alternative (reservoir starvation) fails twice over: the envelope vortex holds *more* total angular momentum, and feeding the fine-grained books shows the boundary-layer import of relative angular momentum is a net *export* until intensification begins in either case. Hugo and Ivan, verified at fixed landfall clocks that fall inside the delay window, read the delay as weakening. There is no free vortex boundary: the taper's cost was aim; the envelope's is a slower Ekman spin-up. We choose the envelope, and we report the price.

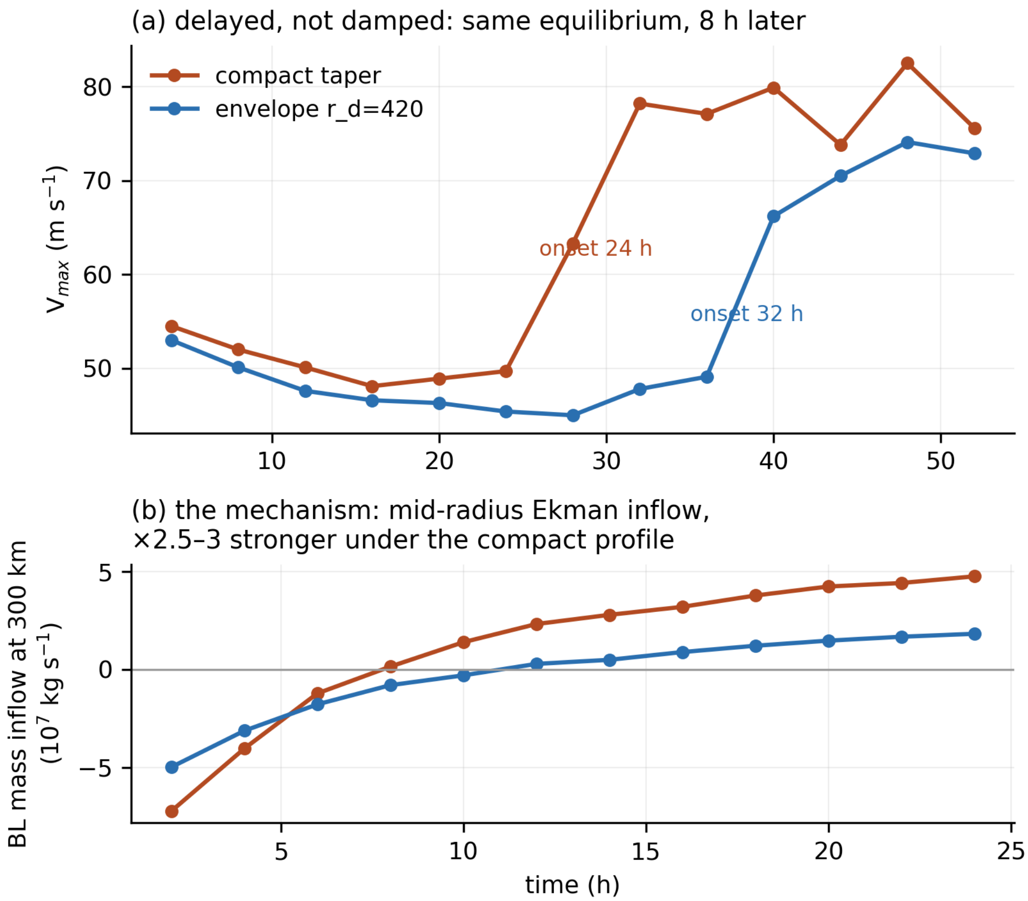

Fig. 4. The envelope's cost. (a) Maximum wind in the emergent-intensification testbed: intensification is delayed (onset 24 h $\rightarrow$ 32 h), not damped—both profiles reach the same capped equilibrium. (b) Boundary-layer mass inflow at 300 km: in the pre-onset window the Ekman inflow runs 2.5–3$\times$ stronger under the compact profile—the mechanism of the delay.

5.4 Three More Storms, Registered Blind

Because six storms carried both the skill claim and the transmission ratio, we extended the A/B by three storms chosen for geometric stress—predictions, guards, and thresholds frozen before any reanalysis was downloaded. Charley (2004) lands moving north-northeast—Michael's axis—and so asks directly whether Michael's under-transmission is a class; its crossing of Cuba, which the model does not represent, was registered in advance as a known unrepresented feature. Florence (2018) lands moving nearly due west, the record's first zonal geometry, for which the westward drift correction should read as *timing* rather than placement. Ida (2021) is a near-pure cross-track case beside the Katrina–Laura falsifier pair. All guards came back clean (arm-to-arm intensity divergence ≤ 2.6 m s^{-1}, timing ≤ 0.8 h), so every number below counts.

Table 3. The three-storm extension: landfall-fix decomposition under control and envelope, the dominant-axis shift, the strong-form projection through the registered heading, and the observed and decomposition-predicted transmission ratios.

Storm	Cross, c → e (km)	Along, c → e (km)	Axis	Obs shift (km)	Strong (km)	Obs ratio	Pred ratio
Charley	−31.8 → −45.0	−160.7 → −172.8	cross	−13.2	−59	0.22	0.42
Florence	+62.7 → +65.7	−124.3 → −78.3	along	+46.0	+73	0.63	0.56
Ida	+36.1 → +13.1	−14.0 → +0.6	cross	−23.0	−97	0.24	0.31

The sign record is now nine for nine: Charley and Ida shift westward, and Florence shifts *forward*—its along-track shift (+46 km) dwarfs its cross-track shift (+3 km), confirming the registered geometric corollary that a zonal mover reads the westward correction as timing. The blind-skill record also extends: every landfall cross-track in the nine-storm record is within 150 km at the fix, and Ida's envelope run lands with 13 km total error, the best in the record. (Charley's placement holds while its timing does not—the model arrives 4–5 h late, unable to follow the observed rapid acceleration and intensification into landfall, with the registered Cuba crossing unrepresented; a dry, capped model's honest limit, visible on the axis the record says it should be visible on.)

The mechanism's first out-of-sample test splits exactly along geometry. The per-storm decomposition, fed the new runs' intensity histories, closes on both quasi-straight movers—Florence within 0.07, Ida within 0.07—and misses Charley by 0.20 (0.42 predicted, 0.22 observed): Michael's shortfall (0.45 predicted, 0.20 observed), replicated under frozen predictions on the record's only other sharply-recurving landfall. The residual is therefore a reproducible class, not a one-off. Two candidate axes describe its membership, and $n = 2$ cannot separate them: both misses curve sharply into landfall, and both carry observed inner cores far smaller than the frozen 75-km initialization—they are the record's two most mis-sized cores. One datum leans against core size alone: Ida's observed radius of maximum wind is 19 km (HURDAT2), as mis-sized as either, and the decomposition closes on it. The decomposition closes on every quasi-straight or zonal mover and fails on both curved landfalls; whether curvature or initialization is doing the work is unresolved here. A post-hoc reading, recorded as a candidate rather than a claim: for a sharply curving track, projecting the transit-accumulated correction through the single final-heading geometry—as both the strong form and the decomposition do—overstates what the correction can deliver, and the registered [0.15, 0.55] transmission band, generalized from cross-axis experience, likewise failed on Florence's along axis (0.63): a long, slow transit spends more of its time at mature drift and transmits harder. Both failures narrow the same thing—the geometry through which a drift correction becomes a landfall displacement—and both are recorded in the ledger.

6 Discussion

One model or a class, sharpened. Any model that bounds its initial vortex with compact support—a common and innocent-looking choice on periodic domains—should check its gyre phase-lock. The diagnostic is an afternoon: quiescent β-plane, balanced vortex, mature-window drift vector, f-plane null; the signature is amplitude saturation with heading precession. The fix costs one profile change and one testbed calibration.

What remains open. The baroclinic nudge's magnitude resists measurement: the natural matched-intensity control—a dry cap-pinned vortex—proved unreadable when run, its track non-steady at cap intensities (the center-finding fragility noted twice above, possibly compounded by genuine trochoidal wobble), so the nudge's size is recorded as unresolved while its sign already exonerates the candidate as cause (Section 2.3). The anchoring window for a held warm core proved narrow—sixfold weaker relaxation ends in the adiabatic runaway—itself a datum on maintaining baroclinic structure in this core. A wobble-robust center-tracker is the prerequisite for any future high-intensity drift measurement. The curved-landfall under-transmission is now a reproducible class of two: Michael (0.20 observed against 0.45 predicted) and, out of sample under frozen predictions, Charley (0.22 against 0.42)—while the decomposition closes on every quasi-straight or zonal mover in the record. The leading untested candidate is the fixed-heading projection itself, which for a sharply curving track overstates what a transit-accumulated correction delivers to the final landfall geometry; a curvature-following projection is a cheap next test. The environmental-sampling bracket has been run. An annulus sweep (5–9° and 7–11° against the production 3–7°) found no robustness to report but something better: the sampled poleward steering drains monotonically with ring radius—for one storm it reverses sign—and at the widest ring two of three storms never reach land. Combined with the structural point that a full-ring vector mean cancels storm-centered azimuthal structure by construction, the verdict is environmental heterogeneity rather than storm contamination: "the environment" is not a scale-free concept for these landfalls, and the production annulus is not one adequate choice among many but essentially the band containing the storm's actual advecting flow—a first-order physical choice, validated by skill. The double-counting suspicion raised in the companion paper is thereby bounded on both grounds, and the obs-anchored test of Section 5.3 completes the pair.

Instrument findings en route. Two byproducts of the budget work merit brief record. The bulk-drag discretization over-counted the column-integrated momentum sink as the vertical grid refines (by 2.1× at doubled resolution; an integral-preserving form is now available, and the production grid's effective drag coefficient is 0.75× its nominal value—a documentation correction, not a behavioral one). And the model's heated secondary circulation is not vertically converged at the production spacing: refinement to and beyond doubled vertical resolution agrees to 0.1 m s^{-1} while the production grid over-produces the heated updraft by $\sim 80\%$—a characterized limitation for future thermodynamically active work (the barotropic track configuration is untouched). Both were found because registered predictions failed and the failures were pursued.

The method, briefly. Thirty-two quantitative predictions were registered across the campaign's seven experiment sets, each committed before the run it concerned; fifteen were confirmed, ten failed, three held in part, three were unscoreable as designed, and one two-branch discriminator resolved against our stated lean (the full ledger accompanies this paper). A subsequent review-hardening round—the persistence baseline, the steering-only tracer, and the per-storm decomposition above—registered eleven further predictions under the same rules (six confirmed, five failed), and the three-storm blind extension of Section 5.4 registered five more (three confirmed, two failed, both failures reported there). Across the program: forty-eight registered predictions, twenty-four confirmed, seventeen failed, seven partial or unscoreable—and the failures, throughout, as instructive as the confirmations. The calibration record is itself informative: the high-confidence bucket verified at seven of seven, while the middle band ran overconfident, its misses concentrated in precisely the two lessons recorded en route—that surface drag couples to intensity only through a driven secondary circulation, and that threshold feedbacks must be registered as trajectories, not endpoints. Most telling, the two deliberate long shots—the 20%-weighted cutoff branch and the 35%-weighted attenuation branch—are this paper's two principal results: the headline findings entered the campaign as its least-favored registered hypotheses. The companion paper argued that a model should never be granted the benefit of the doubt; this paper adds the corollary that neither should the experimenter.

7 Conclusions

The companion paper characterized a bias and declined to explain it; this paper explains it, removes it, and prices both the removal and its side effect. The poleward β-drift aim was the cost of bounding a vortex with compact support: truncation removes the outer flow that phase-locks the β-gyres. A smooth envelope restores canonical, phase-locked, intensity-invariant self-propagation in the testbed, and on nine historical storms—three of them added blind, under predictions registered before their reanalysis was downloaded—moves every landfall the predicted direction—at roughly one-third the projected distance, because a mature-testbed drift overstates what drift does during a real transit (intensity history and gyre spin-up; steering feedback contributes less than a tenth, by direct test, and a vortex-free integration of the steering independently recovers the same footprint). That transmission ratio, measured under registered predictions, is the mechanical content of "subdominant to steering," and it generalizes as a caution that no longer depends on any particular steering architecture: landfall error is a dull instrument for reading self-propagation. The envelope's own cost—an 8–12-h delay in dry re-intensification, traced to weakened mid-radius Ekman inflow—is characterized rather than hidden, in keeping with the discipline both papers exist to demonstrate. The model is better than it was, and we know exactly how much, in which respects, and at what price.

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- [Companion paper, in preparation: "Catching Our Own Errors: Clean Initialization and Blind Track Skill in a Tropical Cyclone Model."]